



Nifty100 Analytics

Analyst Guide

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Project Overview

Nifty100 Analytics is a financial analytics dashboard built using Python, Streamlit, FastAPI, SQLite, and Pandas. The application helps users analyse companies in the Nifty 100 index by presenting financial data through an easy-to-use dashboard. Key Features

- Home dashboard with summary metrics.
- Company Profile with detailed financial information.
- Stock Screener with filtering options.
- Peer Comparison.
- Trend Analysis.
- Sector-wise Analysis.
- Capital Allocation Analysis.
- Reports and downloadable outputs.

Technologies Used

- Python
- Streamlit
- FastAPI
- SQLite
- Pandas
- Git & GitHub

Setup

- Develop a Nifty100 Analytics Dashboard for analysing Nifty 100 companies.
- Provide an interactive Streamlit dashboard for financial analysis.
- Build FastAPI APIs to access company and financial data.
- Store and manage financial data using SQLite.
- Implement ETL processes to import and clean financial datasets.
- Calculate important financial KPIs and ratios.
- Enable company screening based on financial metrics.
- Generate reports and visualisations for analysis.
- Ensure the application is tested, documented, and easy to maintain.
- Provide a fast, reliable, and user-friendly platform for investment analysis.

Dashboard

← ↻ ⓘ localhost:8501

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Nifty100 Analytics

Navigation

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📄 Company Profile

📊 Screener

👥 Peers

📈 Trends

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📄 Reports

 **Home**

Welcome to the Nifty100 Analytics Dashboard!

Show desktop

<<

app

home

profile

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reports

Select Year

2024

▼

Deploy ⋮

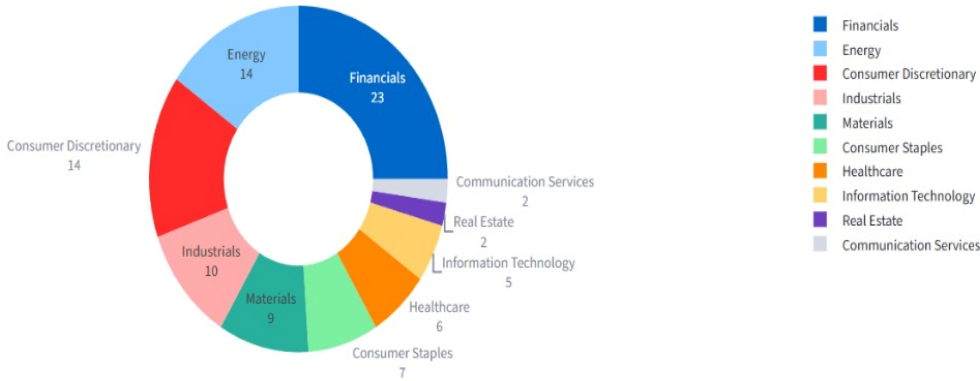


Home

Average ROE	Median P/E	Median D/E	Total Companies	Median Revenue CAGR (5Y)	Debt-Free Companies
114.29%	46.18	0.45	92	-	0

Sector Breakdown

Companies by Sector





Top 5 Companies by Composite Quality Score

Company	Sector	Quality Score
Bharat Electronics Ltd	Industrials	64.12
Bajaj Holdings & Investment Ltd	Financials	63.66
Hindustan Aeronautics Ltd	Industrials	58.89
Tata Consultancy Services Ltd	Information Technology	41.82
Interglobe Aviation Ltd	Consumer Discretionary	41.33

Company Profile

Search by company name or ticker

HDFC

Press Enter to apply

Matching companies

HDFC Bank Ltd (HDFCBANK)

HDFC Bank Ltd

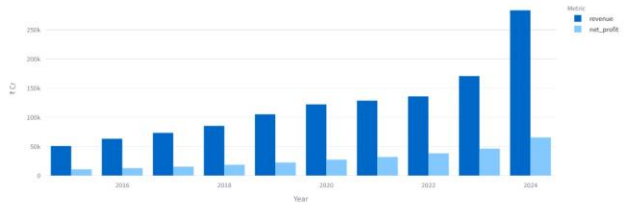
Sector: Financials Sub-Sector: Private Banks NSE Ticker: HDFCBANK

HDFC Bank Limited (also known as HDFC) is an Indian banking and financial services company headquartered in Mumbai. It is India's largest private sector bank by assets and the world's tenth-largest bank by market capitalization as of May 2024. As of April 2024, HDFC Bank has a market capitalization of \$145 billion, making it the third-largest company on the Indian stock exchanges

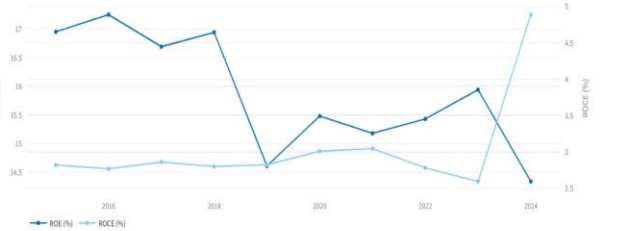
ROE	ROCE	Net Profit Margin	D/E	Revenue CAGR (5yr)	FCF (₹ Cr)
14.34%	7.67%	23.07%	6.81	21.95%	35669.00

Revenue & Net Profit (10-Year)

Revenue & Net Profit (10-Year)



ROE & ROCE (10-Year)



Pros & Cons

- Pros**

 - Company is expected to give good quarter
 - Company has delivered good profit growth of 23.4% CAGR over last 5 years
 - Company has been maintaining a healthy dividend payout of 22.9%
 - Company's median sales growth is 16.4% of last 10 years
- Cons**

 - Stock is trading at 2.76 times its book value
 - Company has low interest coverage ratio.
 - Contingent liabilities of Rs.24,09,821 Cr.
 - Earnings include an other income of Rs.1,55,870 Cr.

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Select Year
2024

Filters
ROE Min
15.00
Debt/Equity Max
2.00
PE Max
50.00

Deploy

Nifty100 Stock Screener

Companies Found: 31

	id_x	company_id	year	net_profit_margin_pct	operating_profit_margin_pct	return_on_equity_pct	debt_to_equity	interest_coverage	asset_turnover	
5	66	ADANIPOWER	2024	30.34		58	15.35	0.9373	5.9495	0.2283
6	78	ADANIPOWER	2024	30.34		58	15.35	0.9373	5.9495	0.2283
7	90	ADANIPOWER	2024	41.37		36	48.28	0.8023	8.2972	0.5472
9	113	APOLLOHOSP	2024	4.91		13	13.48	0.7689	5.5657	1.1384
12	160	BAJAJ-AUTO	2024	17.18		20	26.61	0.066	174.4367	1.1405
13	161	BAJAJ-AUTO	2024	17.18		20	26.61	0.066	174.4367	1.1405
16	197	BAJAJHLDNG	2024	432.73		92	13.58	0.0012	3764.5	0.0261
28	339	COALINDIA	2024	26.26		47971	45.17	0.0788	11.2418	0.6019
33	399	DRREDDY	2024	19.91		28	19.74	0.0709	51.7076	0.7223
34	411	EICHERMOT	2024	24.2		26	22.17	0.0232	114.7059	0.7154

Download CSV

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Select Company
Abbott India Ltd

Select Metrics (Max 3)
Revenue x EPS x Net Margin x

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Trend Analysis

10-Year Financial Trend

Value

6000
5000
4000
3000
2000
1000

3.1%
7.1%

Revenue
EPS
Net Margin

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Select Year
2024

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Peer Comparison

Select Peer Group
Automobiles

Peers Group: Automobiles

	company_name	return_on_equity_pct	net_profit_margin_pct	debt_to_equity	free_cash_flow_cr	pe_ratio	pb_ratio
23	Maruti Suzuki India Ltd	15.75	9.51	0.0024	4936	20.78	8.26
24	Tata Motors Ltd	37.46	7.26	1.2031	45133	67.07	0.93
25	Mahindra & Mahindra Ltd	18.54	8.82	1.6414	-11245	14.22	0.81
26	Bajaj Auto Ltd	26.61	17.18	0.066	6486	29.91	9.83
27	Bajaj Auto Ltd	26.61	17.18	0.066	6486	29.91	9.83
28	Eicher Motors Ltd	22.17	24.2	0.0232	890	15.62	13.61
29	Hiero MotoCorp Ltd	21.14	9.9	0.0342	3095	53.86	12.6
30	TVS Motor Company Ltd	26.22	4.54	3.8334	-2254	46.1	5.62

Download CSV

Deploy

Financial Data

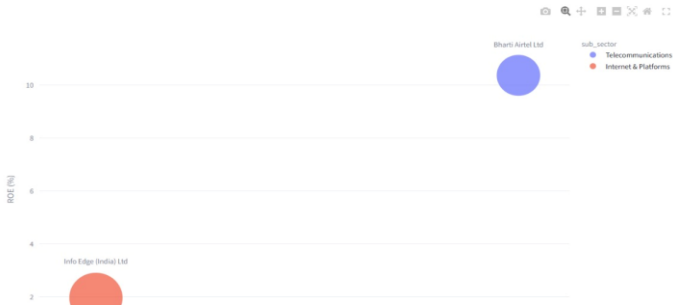
	year	Revenue	Net Profit	EPS	ROE	Net Margin	Debt to Equity	Free Cash Flow
0	2012	1653	145	68	22.41	8.77	0	42
1	2014	2276	198	93	25.13	8.7	0	11
2	2014	2276	198	93	25.13	8.7	0	0
3	2015	2289	229	108	24.44	10	0	28
4	2015	2289	229	108	24.44	10	0	-1899
5	2016	2614	255	120	21.34	9.76	0	172
6	2016	2614	255	120	21.34	9.76	0	728
7	2017	2903	277	130	19.97	9.54	0	152
8	2017	2903	277	130	19.97	9.54	0	459
9	2018	3298	401	189	23.69	12.16	0	-42

Sector Analysis

Select Sector

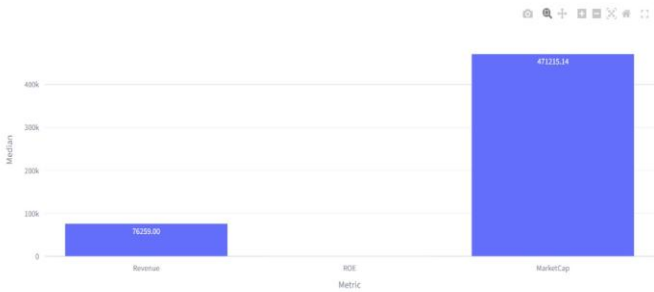
Communication Services

Sector Bubble Chart



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Sector Median KPIs



Companies

	company_name	Broad_sector	sub_sector	year	Revenue	ROE	MarketCap
0	Bharti Airtel Ltd	Communication Services	Telecommunications	2024	149982	10.36	378605.83
1	Info Edge (India) Ltd	Communication Services	Internet & Platforms	2024	2536	1.97	563824.44

Capital Allocation Map



Companies

Select Capital Pattern

High ROE

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Reports

Company Report

Search Company

	id	company_name
0	ABB	Abbott India Ltd
1	ADANIENSOL	Adani Energy Solutions Ltd
2	ADANIENT	Adani Enterprises Ltd
3	ADANIGREEN	Adani Green Energy Ltd
4	ADANI PORTS	Adani Ports & Special Economic Zone Ltd
5	ADANIPOWER	Adani Power Ltd
6	ATGL	Adani Total Gas Ltd
7	AMBUJACEM	Ambuja Cements Ltd
8	APOLLOHOSP	Apollo Hospitals Chain of Indian private hospitals
9	ASIANPAINT	Asian Paints Indian Multi-National Paint and Coating Manufacturing Company

Download Report (CSV)

Deploy

Company Profile

- Search company and view KPIs.
- Display company name, sector and industry.
- View financial ratios (ROE, ROCE, Debt-to-Equity, P/E).
- Analyse revenue, profit and growth trends.
- Compare company performance over multiple years.
- Review company strengths and weaknesses.
- Support investment analysis and decision-making.

Screenener

- Apply financial filters to screen companies.
- Filter by ROE, ROCE, P/E, Debt-to-Equity and Market Capitalisation.
- View matching companies instantly.
- Compare financial performance across companies.
- Export screening results for further analysis.
- Support investment research and decision-making.

API Usage

- Example: GET <http://localhost:8000/docs>



Troubleshooting

- Port conflicts (8501/8000 already in use)
- Incorrect database path (db/nifty100.db)
- Missing Python packages (pip install -r requirements.txt)
- API server not running
- Empty database or missing tables
- Streamlit page not loading
- Check terminal logs for error messages
- Restart Streamlit after code changes

Conclusion

- Successfully developed the Nifty100 Analytics Dashboard.
- Integrated SQLite, Streamlit, and FastAPI.
- Supports company analysis, screening, trends, sectors, peers, capital, and reports.
- Follow the setup steps in the README before running the project.
- Ensure the database is loaded and API is running for all features.
- The project is ready for analysis, testing, and future enhancements.